Project Synopsis

1. Title: Smart Human – Computer Interaction Using Hand Gestures

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
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Program Track and Domain

Program Track	Domain
Engg-Spoke	Machine Learning, Computer Vision

Guide Details

Guide Name: Rapaka Manoj Kumar

Designation: SME

2. Problem Statement

Traditional Human-Computer Interaction (HCI) systems rely heavily on physical devices such as a mouse, keyboard, or touchpad. These devices limit natural interaction and may not be convenient in situations where touchless control is preferred, such as healthcare environments, presentations, or assistive technologies.

The problem addressed in this project is the dependency on physical input devices for controlling computer systems. There is a need for a touchless, intuitive, and real-time interaction system that enables users to control computer functions using natural hand gestures.

This project proposes a Smart Human-Computer Interaction system using Hand Gestures, where users can control mouse movements and actions such as clicking, scrolling, and dragging using only their hand gestures captured through a webcam. The system leverages computer vision and machine learning techniques to detect and track hand landmarks and map them to corresponding mouse actions.

3. Objectives

The specific objectives of this project are:

• To develop a real-time hand gesture recognition system using a webcam.

• To detect and track hand landmarks accurately.

• To map specific hand gestures to mouse actions such as:

- Cursor movement
- Left click
- Right click
- Double click
- Scrolling

• To achieve smooth and stable cursor control.

• To design a low-cost and hardware-independent solution using only a standard webcam.

4. Implementation Plan

The project is implemented using a structured workflow consisting of image capture, hand detection, gesture recognition, and mouse control mapping.

Step 1: Video Capture

The system captures live video input from the webcam using OpenCV.

Step 2: Hand Detection

Hand detection and tracking are performed using:

- MediaPipe
- OpenCV

MediaPipe provides 21 hand landmarks that help identify finger positions and movements.

Step 3: Landmark Extraction

The coordinates of finger tips (especially index and middle finger) are extracted and used to calculate distances between fingers.

Step 4: Gesture Recognition Logic

Different gestures are defined based on finger positions:

Gesture	Finger Position	Action Performed
☝ Index Finger Only	Only index finger open	Move Cursor
🤟 Index + Middle + Ring fingers	Distance between index, middle & ring fingers are small	Left Click
🤏 Hold Pinch	Pinch and hold	Drag
✊ Fist	All fingers close	Release (Stop Drag)
🖐 Four fingers	Thumb finger close & other are closed	Scroll
✌ + Swipe Left/Right	Two fingers + horizontal motion	Next / Previous Tab`
✌ Index + Middle Open	(Hold) Two fingers open	Copy
🤟 Index + Middle + Ring fingers	(Hold) Distance between index, middle & ring fingers are small	Paste
🤙 Thumb + Little Finger	Only thumb & little finger open	Exit Application

Step 5: Mouse Control

Mouse operations are performed using:

- PyAutoGUI

The recognized gesture is converted into corresponding mouse commands.

Step 6: Optimization

To avoid shaky cursor movement:

- Frame smoothing techniques are applied
- Distance thresholds are defined
- Sensitivity scaling is implemented

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware Requirements

• Laptop/PC with minimum Intel i5 / Ryzen 5
• Minimum 8GB RAM
• Built-in or external webcam
• No GPU required (optional)

Software Requirements

• Operating System: Windows / Linux / macOS

• Programming Language: Python

• Development Environment: Jupyter Notebook / VS Code / IDLE

• Libraries:

- OpenCV
- MediaPipe
- PyAutoGUI
- NumPy

B. Dataset

This project does not require a pre-labeled dataset.

• Data Type: Real-time video from webcam

• Data Source: Live camera feed

• Processing Type: Real-time landmark detection

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Problem Definition & Research	Day 1 – Day 2
Phase 2	System Design	Day 3 – Day 4
Phase 3	Hand Detection Implementation	Day 5 – Day 7
Phase 4	Gesture Mapping & Mouse Control	Day 8 – Day 10
Phase 5	Testing & Optimization	Day 11 – Day 13
Phase 6	Documentation & Final Report	Day 14 – Day 15

6. Expected Outcomes

By the completion of this project, the following outcomes are expected:

- A fully functional virtual mouse controlled by hand gestures
- Real-time and smooth cursor movement
- Accurate gesture recognition with minimal delay
- Reduced dependency on physical mouse devices
- A practical demonstration of Human-Computer Interaction using AI

7. References

- https://opencv.org/
- https://developers.google.com/mediapipe
- https://pyautogui.readthedocs.io
- Research papers on Hand Gesture Recognition & HCI